

CTF-GS: Compact Foundation-Feature Fields for Open-Vocabulary 3D Gaussian Scene Understanding

Anonymous Authors

Abstract—Dense vision foundation models provide strong 2D features, but deployed 3D scenes still lack a compact representation that can reuse those features for open-vocabulary queries from novel views and directly in 3D. We present CTF-GS, a compact foundation-feature field for 3D Gaussian scenes. Rather than storing raw high-dimensional teacher features on every Gaussian or training a scene-specific classifier, CTF-GS distills frozen RADIO features into a Contextual Gaussian Feature Field, reconstructs foundation-space features through a Foundation-Space Reconstruction decoder, and exposes feature quality and visibility as label-free field outputs. The same scene memory supports complementary readouts: rendered dense feature maps for novel-view localization, direct Gaussian primitive scores for 3D object selection, and compact point features for cross-dataset semantic querying. Multiview Primitive Registration provides an auditable multiview bridge for transferring registered view evidence into the compact direct field during training.

We evaluate CTF-GS under three protocols. On LERF-OVS rendered-view grounding, CTF-GS reaches 0.8598 macro localization accuracy and 0.5889 macro mIoU. Under an OpenGaussian-style direct 3D object-selection protocol, a compact prompt-ensemble readout with label-free color-edge support calibration reaches 0.5014 macro mIoU and 0.7044 Acc@0.25 without reading a multiview-registration cache or invoking an official RGB SAM decoder at inference. On the VALA/OpenGaFF ScanNet-8 direct point-query protocol, the compact field reaches 0.3806, 0.3871, and 0.4711 mIoU on the 19-, 15-, and 10-class splits. Same-evaluator frame-wise foundation-feature comparisons and component ablations show that multiview reconstruction and support-calibrated direct 3D readout drive the gains. Small fragmented objects remain the primary failure mode.

Index Terms—Open-vocabulary 3D scene understanding, 3D Gaussian Splatting, vision foundation models, feature distillation, scene representation.

I. INTRODUCTION

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding, this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views, select objects directly in 3D, and transfer to new point-query protocols without storing a dense teacher feature on every primitive.

CTF-GS targets this gap by reconstructing foundation-model features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether a scene representation can support multiple interfaces

from the same compact field: dense novel-view feature maps for rendered open-vocabulary localization, direct primitive scores for 3D object selection, and point-level features for semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Contextual Gaussian Feature Field, reconstruct foundation-compatible features through a Foundation-Space Reconstruction module, and calibrate them in view space before comparison to the frozen RADIO teacher. We also use Frozen-Head Geometry Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

We evaluate this formulation with a protocol boundary that mirrors recent open-vocabulary Gaussian-field papers. LERF-OVS rendered-view grounding tests whether the field can produce dense novel-view features; LERF direct 3D object selection tests whether text queries can select Gaussian primitives before any rendered-mask evaluation; and VALA/OpenGaFF-8 ScanNet direct point querying tests whether the same compact representation remains usable outside the LERF object-query setting. External results are used as source-anchored context when they are not rerun under the same evaluator, while method claims are tied to fixed evidence files and same-protocol ablations.

a) Contributions.: We make four contributions:

- We formulate compact foundation-feature reconstruction for queryable 3D Gaussian scene understanding across rendered-view localization, direct Gaussian object selection, and direct point queries.
- We introduce a unified multi-head Contextual Gaussian Feature Field that stores compact per-Gaussian codes, augments them with spatial context, and predicts feature quality and visibility without direct high-dimensional foundation-feature storage.
- We use Foundation-Space Reconstruction, View-Conditioned Feature Calibration, Frozen-Head Geometry Consistency, and multiview-registration-to-field transfer to recover teacher-compatible features while keeping teachers and downstream heads fixed; multiview registration acts as a registered supervision bridge rather than a required inference cache for the compact direct readout.
- We provide a protocol-separated evaluation covering LERF-OVS rendered grounding, OpenGaussian-style di-

rect 3D object selection, ScanNet direct point-query transfer, qualitative overlays, storage, runtime, and failure analysis.

II. RELATED WORK

a) *Language fields and open-vocabulary 3D understanding.*: LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [1]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [2], [3]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [4]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [5], while InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [6], [7]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [8], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [9]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [10]. CTF-GS differs in its emphasis on compactly reconstructing a high-dimensional unified teacher feature space and on compressing registered multiview evidence back into a compact primitive-level readout.

b) *Gaussian scene representations.*: 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [11]. CTF-GS keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

c) *Agglomerative vision foundation models.*: RADIO-style models distill multiple visual capabilities into a single dense vision backbone [12], [13]. CTF-GS uses C-RADIOv4-H as a frozen teacher and studies whether its dense features can be reconstructed from 3D scenes at novel views.

d) *Foundation-model regularization in 3DGS.*: FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [14]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors into 3D Gaussian features for promptable object segmentation [15]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

III. METHOD

A. Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a compact feature field that is compatible with a frozen RADIO teacher and can be queried through both rendered views and 3D locations. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The rendered teacher target is:

$$F_v^* = T(I_v) \in \mathbb{R}^{H \times W \times 1280}. \quad (1)$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against F_v^* . The same compact field can also be decoded at Gaussian centers or query points, which makes direct primitive and point-level readouts part of the representation rather than separate task-specific classifiers.

B. Contextual Gaussian Feature Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. CTF-GS attaches compact learnable latent features to the Gaussians and combines two paths:

- 1) a fine path that splats per-Gaussian latent features into the image plane;
- 2) a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused by decoupled geometry and semantic heads into a compact feature map Z_v . In the unified multi-head field, the same fused state also predicts a feature-quality logit q_v and a visibility logit u_v :

$$(Z_v, q_v, u_v) = H_\phi(Z_v^{fine}, Z_v^{coarse}, d_v, \alpha_v). \quad (2)$$

The quality head estimates whether a rendered compact feature will reconstruct the frozen teacher reliably, while the visibility head estimates whether the rendered evidence is supported by visible Gaussian mass. Both heads are available for rendered-view readout and direct Gaussian/point readout. A Foundation-Space Reconstruction module maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(Z_v). \quad (3)$$

A View-Conditioned Feature Calibration module then improves local alignment using rendered guidance signals such as visibility, alpha, and geometric depth.

C. Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^* \rangle}{\|\hat{F}_v\|_2 \|F_v^*\|_2} + \lambda_{rec} \|\hat{F}_v - F_v^*\|_1. \quad (4)$$

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc} \mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \quad (5)$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

The unified quality heads add two label-free auxiliary objectives:

$$y_v^q = \frac{1 + \cos(\hat{F}_v, F_v^*)}{2}, \quad y_v^u = \alpha_v, \quad (6)$$

where gradients are stopped through the targets. We train q_v and u_v with binary cross-entropy against y_v^q and y_v^u respectively. These losses do not introduce task labels; they expose feature reliability and visibility as first-class field outputs that can be reused by text grounding, direct primitive scoring, and failure analysis.

CTF-GS: one compact foundation-feature Gaussian map

Frozen foundation heads supervise training; deployed readouts query the same compact scene memory as 2D features, 3D primitives, and 3D points.

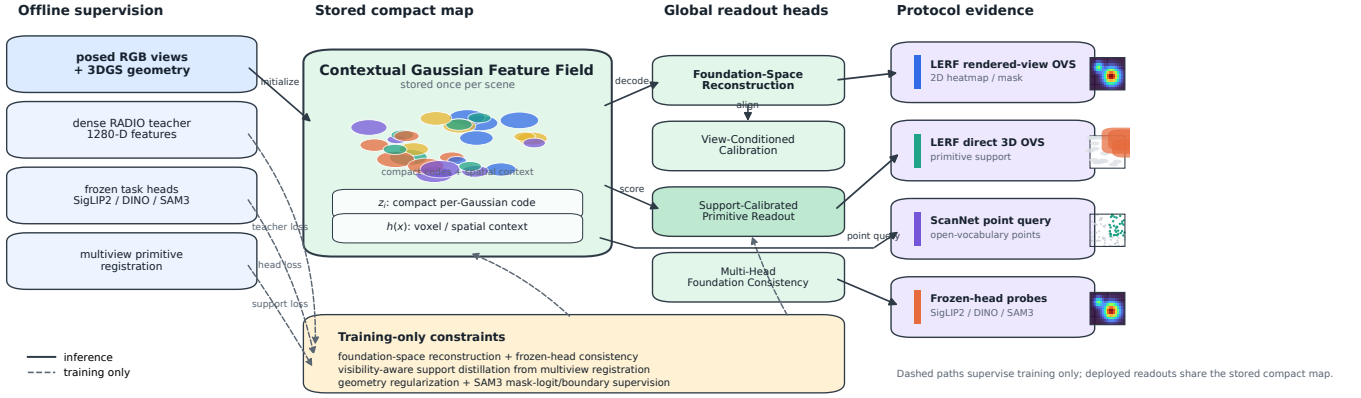


Fig. 1. CTF-GS framework. The method stores one compact foundation-feature Gaussian map with per-Gaussian codes, spatial context, and quality/visibility heads, then decodes it through shared global readout heads. The same map supports rendered dense feature maps for LERF 2D open-vocabulary grounding, compact primitive scores with a support-calibrated primitive readout for LERF direct 3D object selection, and direct point features for ScanNet queries. Frozen RADIO/SigLIP2/DINOv3/SAM3 heads, multiview-registration-to-field support distillation, and geometry/visibility losses are training constraints; the main compact direct readout does not require a multiview-registration feature cache or an official RGB SAM decoder.

D. Multi-Head Foundation Consistency

C-RADIOv4-H exposes frozen adaptor heads for siglip2-g, dino_v3, and sam3. The main open-vocabulary evaluator uses the text-aligned siglip2-g path, while training can additionally supervise the rendered feature through the other adaptors. For an adaptor A_a , we add a pointwise consistency term:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^*)). \quad (7)$$

For dino_v3, we also match the token self-similarity structure rather than only the local feature value. Given normalized adaptor tokens D and D^* , we compute:

$$\mathcal{L}_{rel} = \|DD^\top - D^*D^{*\top}\|_2^2. \quad (8)$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. We also evaluate a cross-view variant inspired by ProFuse’s use of dense cross-view correspondence and context fusion [16]. For two training views p and q , we match teacher and rendered DINO token affinity:

$$\mathcal{L}_{cv} = \|D_p D_q^\top - D_p^* D_q^{*\top}\|_2^2. \quad (9)$$

Unlike ProFuse, this is an optimization-time regularizer on rendered feature maps, not a full pre-registration or context-proposal pipeline. To prevent cross-view smoothing from erasing text-grounding peaks, we add a label-free SigLIP text-heatmap distillation term over a fixed LERF text bank. The per-pixel query variant preserves the scene-category distribution:

$$\mathcal{L}_{txt}^{query} = \text{KL}(\text{softmax}(S^* T^\top / \tau) \parallel \text{softmax}(\hat{S} T^\top / \tau)). \quad (10)$$

We also evaluate a spatial variant that normalizes each query over image locations, directly targeting the LERF argmax localization failure mode:

$$\mathcal{L}_{txt}^{spatial} = \text{KL}(\text{softmax}_{xy}(s_q^* / \tau) \parallel \text{softmax}_{xy}(\hat{s}_q / \tau)). \quad (11)$$

For sam3, the training branch uses a mask-free soft-region loss. Teacher SAM3 tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by teacher similarity, and the rendered feature is trained to match the teacher region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\text{sam3}}(\hat{F}_v)) - \mu_k(A_{\text{sam3}}(F_v^*))\|_2^2. \quad (12)$$

This approximates SAM-style region supervision without requiring external SAM mask caches, while keeping the inference-time representation unchanged. We additionally implement a mask-logit distillation fallback for the sam3 adaptor. Teacher SAM3-adaptor tokens define deterministic mask anchors, and we match the rendered feature’s per-token soft assignment logits to the teacher distribution:

$$\mathcal{L}_{mask} = \text{KL}\left(\text{softmax}(A_{\text{sam3}}(F_v^*) P^\top / \tau) \parallel \text{softmax}(A_{\text{sam3}}(\hat{F}_v) P^\top / \tau)\right), \quad (13)$$

where P are teacher anchor tokens. This remains an adaptor-space training loss rather than an official frozen SAM3 mask-decoder loss. The official SAM3 weights are used only in the separate promptable readout/cache path, where the decoder is frozen and does not train the evaluator.

E. Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

F. Inference Readouts

At evaluation time, CTF-GS renders a feature map for each novel view. For a text query q , we compute a text embedding e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ . The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center readout decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_\theta(c_i), \quad s_i(q) = \cos(A_{\text{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This readout is structurally the same compact field used by ScanNet point-query experiments. For LERF primitive selection, we use a frozen SigLIP2 prompt ensemble and a Support-Calibrated Primitive Readout: selected Gaussians are rendered to a coarse mask, label-free color-edge support calibration refines local boundaries, and a component guard keeps the largest component only when it dominates the refined support; otherwise it preserves multi-component support only above a fixed support-area floor. This is a label-free readout rule. It uses the evaluation RGB image only through a classical color-edge support refinement, not through a learned feature extractor or official SAM decoder, and uses LERF masks only for the final query-select-render evaluation.

We also retain *Multiview Primitive Registration* as a supervision bridge. It renders teacher-space features from posed views, applies view-conditioned calibration and the frozen SigLIP2 head, registers those text-aligned features back to visible Gaussian centers with depth/alpha checks, and then runs the same primitive-level text query on the registered Gaussian embeddings. To make the direct compact field itself more usable, we additionally train a label-free multiview-registration-to-field consistency variant. Let $\bar{e}_i^{\text{mpr}}$ be the cached normalized summary feature registered to Gaussian i , and let n_i be the number of posed views that registered valid evidence to Gaussian i . The transfer objective uses a label-free confidence weight $w_i = \log(1 + n_i) / \text{mean}_j \log(1 + n_j)$ over valid registered primitives. We fine-tune the compact field and a lightweight summary adaptor with

$$\mathcal{L}_{v2f} = 1 - \cos(A_{\text{SigLIP2}}(D_\theta(c_i)), \text{sg}[\bar{e}_i^{\text{mpr}}]) + \lambda \mathcal{L}_{\text{text}},$$

where $\mathcal{L}_{\text{text}}$ matches the registered multiview scene-query distribution and uses only the LERF text categories, not object masks; all loss terms are averaged with w_i . This stage tests whether registered multiview evidence can be compressed back into the queryable Gaussian-center field. The main compact readout does not read the registration cache at inference; the frozen SAM3 box-prompt boundary readout described below is reported separately as a post-selection boundary readout rather than the main compact-only claim. For the Dr. Splat-inspired audit, we also implement rasterizer-level Gaussian-pixel hit registration variants: all footprint contributions, per-pixel dominant hits, and per-Gaussian top-contribution pixels. These variants capture real rasterization intersections instead of only projected centers, but they are treated as negative controls because they reduce the LERF direct-3D metrics under the fixed protocol.

TABLE I

FROZEN CTF-GS LERF-OVS RENDERED-FEATURE GROUNDING RESULT. MIOU USES A LABEL-FREE PEAK-RELATIVE MASK THRESHOLD OF 0.60, A PEAK-CONNECTED-COMPONENT COARSE MASK, AND A FEATURE-ONLY PROMPT-CONDITIONED SAM3 BOUNDARY READOUT WITH HEATMAP-SUPPORT ACCEPTANCE. THE BOUNDARY READOUT RECEIVES RENDERED CTF-GS FEATURES, THE TEXT PROMPT, AND THE COARSE MASK; IT DOES NOT CALL RGB SAM3 AT EVALUATION TIME. LOCACC IS MEASURED FROM THE HEATMAP ARGMAX BEFORE MASK EXTRACTION.

Scene	LocAcc	mIoU	Temp.
Figurines	0.8214	0.5243	50
Ramen	0.9014	0.6325	40
Teatime	0.8983	0.6515	25
Waldo Kitchen	0.8182	0.5475	25
Macro	0.8598	0.5889	–

TABLE II

LERF-OVS OPEN-VOCABULARY GROUNDING COMPARISON. EXTERNAL ROWS ARE OFFICIAL-SOURCE LOCACC VALUES CONVERTED TO DECIMALS; CTF-GS IS THE LOCAL RENDERED-FEATURE RESULT. BEST PER COLUMN IS IN **BOLD**.

Method	Fig.	Ramen	Tea.	Waldo	Macro
LERF	0.795	0.625	0.938	0.815	0.793
LangSplat	0.804	0.732	0.881	0.955	0.843
LEGaussians	0.767	0.737	0.683	0.523	0.678
CTF-GS	0.821	0.901	0.898	0.818	0.860

IV. EXPERIMENTS

A. Evaluation Protocol and Provenance

All CTF-GS numbers are tied to fixed evaluation outputs and a provenance manifest. The rendered-view and direct-3D LERF runs use the same four annotated LERF-OVS scenes; the ScanNet study uses the VALA/OpenGaff-8 scene subset and the class splits described below. External baseline results are official-source values from the cited papers or supplements and are captioned as cross-paper context rather than reproduced same-evaluator results; the exact source results and protocol notes are recorded in `baseline_source_verification.md`.

B. LERF-OVS Main Result

Table I reports the rendered-view grounding result. The readout computes the same rendered SigLIP2 heatmap as the localization metric, removes disconnected thresholded regions that do not contain the query peak, and uses the remaining mask as a coarse prompt for a feature-only SAM3-adaptor boundary head trained from official SAM3 pseudo masks on unlabelled training views. A refined mask is accepted only if it preserves the query peak and enough normalized heatmap mass, so the module can sharpen boundaries without moving away from the rendered feature response. Relative to the peak-component core readout, macro mIoU rises from 0.5707 to 0.5889 with unchanged LocAcc.

Table II provides the external comparison. The LERF, LangSplat, and LEGaussians results are traced to official paper or supplement tables and converted to decimal LocAcc when originally reported as percentages. Because these results are not rerun under the RADIO-GS evaluator, we use them as

TABLE III

LERF-OVS DIRECT 3D OBJECT SELECTION UNDER AN OPENGAUSSIAN-STYLE QUERY-SELECT-RENDER PROTOCOL. EXTERNAL VALUES ARE OPENGAUSSIAN OFFICIAL PAPER NUMBERS. THE REGISTERED MULTIVIEW RESULT USES PRIMITIVE FEATURES TRANSFERRED FROM POSED RENDERED VIEWS. THE COMPACT-FIELD RESULT USES GAUSSIAN-CENTER PRIMITIVE SCORES WITH AN OPACITY-GATED POINT ADAPTER, A FROZEN SIGLIP2 PROMPT ENSEMBLE, AND LABEL-FREE COLOR-EDGE/SCORE-COMPONENT SUPPORT CALIBRATION; IT DOES NOT READ A MULTIVIEW-REGISTRATION CACHE OR OFFICIAL RGB SAM READOUT AT INFERENCE, BUT IT IS NOT THE STRICT NO-RGB ONE-MAP ABLATION. THE SAM3-BOX RESULT IS A FROZEN BOUNDARY READOUT AFTER COMPACT 3D PRIMITIVE SELECTION.

Method	Protocol	Fig.	Ramen	Tea.	Waldo	Macro
OpenGaussian	mIoU	0.393	0.310	0.604	0.227	0.384
CTF-GS MPR	thrOp25 mIoU	0.531	0.580	0.566	0.243	0.480
CTF-GS compact	score-component mIoU	0.510	0.600	0.564	0.331	0.501
CTF-GS	SAM3 box fixed mIoU	0.614	0.641	0.613	0.414	0.570
OpenGaussian	Acc@0.25	0.554	0.422	0.763	0.318	0.514
CTF-GS MPR	thrOp25 Acc@0.25	0.786	0.746	0.763	0.409	0.676
CTF-GS compact	score-component Acc@0.25	0.679	0.831	0.763	0.545	0.704
CTF-GS	SAM3 box fixed Acc@0.25	0.696	0.746	0.746	0.546	0.684

TABLE IV

COMPACT DIRECT3D READOUT ABLATION. ALL VARIANTS USE COMPACT GAUSSIAN-CENTER PRIMITIVE SCORES AND THE OPENGAUSSIAN-STYLE QUERY-SELECT-RENDER EVALUATION. THE PURE ONE-MAP VARIANT IS THE STRICT NO-REGISTRATION-CACHE/NO-RGB/NO-SAM READOUT; GUARDED VARIANTS ADD LABEL-FREE COMPONENT SUPPORT CALIBRATION BUT STILL AVOID A MULTIVIEW-REGISTRATION CACHE AND OFFICIAL RGB SAM DECODER.

Readout	MPR cache	SAM3	RGB	Prompt	mIoU	Acc@0.25
Compact single prompt, pure one-map	no	no	no	no	0.449	0.672
Compact direct, base guarded readout	no	no	yes	no	0.484	0.643
Compact prompt ensemble, pure one-map	no	no	no	yes	0.457	0.685
Compact prompt ensemble + RGB component guard	no	no	yes	yes	0.500	0.705
Compact prompt ensemble + RGB/score-component guard	no	no	yes	yes	0.501	0.704

context for the main claim rather than as a strict reproduced leaderboard.

Supplementary calibration isolates the rendered-mask boundary readout from feature quality. Raising the label-free peak-relative threshold from 0.50 to 0.60 tightens masks and improves macro mIoU from 0.494 to 0.524, while peak-connected-component filtering lifts the same four-scene readout to 0.5707. The feature-only SAM3-adaptor boundary head is therefore used only after the peak component has supplied a stable coarse support, turning it into a boundary refinement step rather than a new localization mechanism.

C. LERF Direct 3D Object Selection

Table III evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table I; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

Table IV separates primitive scoring from support selection. A strict no-prompt/no-RGB compact one-map readout reaches 0.449 macro mIoU and 0.672 Acc@0.25. Adding only the frozen prompt ensemble raises this to 0.457/0.685, while label-free color-edge and score-component support calibration

TABLE V

PUBLISHED-CONTEXT LERF-OVS DIRECT 3D OBJECT-SELECTION NUMBERS. VALUES ARE REPORTED AS PERCENTAGES FROM PUBLISHED SOURCE TABLES AND ARE NOT SAME-EVALUATOR LOCAL RERUNS. RESULTS FOLLOW THE RECENT OPENGAFf COMPARISON SET, WITH THE OPENGAFf METHOD INTENTIONALLY OMITTED.

Method	Source type	mIoU	Acc@0.25
OpenGaussian [4]	official paper	38.36	51.43
Dr. Splat [5]	published context	43.29	64.30
InstanceGaussian [6]	published context	45.30	58.44
CTF-GS + MPR	local, SigLIP2, fixed thrOp25 + color-edge calibration	48.01	67.60
CTF-GS compact direct field	local, no MPR/SAM cache, score-component guard	50.14	70.44
CTF-GS + SAM3 box	local boundary readout, frozen official SAM3	57.05	68.35

reaches 0.501/0.704. The gain is largest on Waldo Kitchen, where Acc@0.25 rises from 0.409 for the earlier compact readout to 0.545 by recovering small or fragmented object support. This makes the deployed compact direct readout stronger than the registered multiview readout in Acc@0.25 without reading a multiview-registration feature cache or using an official SAM3 mask decoder.

The frozen official SAM3 box-prompt result is reported separately as a boundary readout after compact 3D selection. It reaches 0.570 macro mIoU and 0.684 Acc@0.25, but SAM3 is invoked only after primitives have already been scored and rendered; its candidate mask is selected by overlap with the rendered prediction rather than ground truth. This result is useful for measuring the remaining boundary gap, but the core direct-field claim is carried by the compact readout. Waldo Kitchen still shows that instance grouping can help fragmented objects, so we treat these results as evidence for direct-field usability rather than as a universal primitive-level ranking claim.

Table V gives the published-context baseline results used by the recent OpenGaFF paper, with the OpenGaFF method result omitted by design. These results are deliberately separated from the local result table: they mix papers, text heads, and implementation details, so they motivate positioning against recent primitive-/instance-aware methods rather than serving as a strict same-evaluator leaderboard.

Supplementary protocol audits verify that Multiview Primitive Registration uses 128 posed rendered views, visibility checks, and no LERF masks or query labels for registration. They also record negative or auxiliary variants such as fixed-ratio selectors, best-by-scene thresholds, contribution-weighted raster registration, proposal/OPR component selection, and confidence selectors. The main conclusion is that feature registration improves primitive evidence, but stable object selection requires Support-Calibrated Primitive Readout; query-level audits identify Waldo Kitchen as a coverage, ambiguity, and boundary-fragmentation bottleneck.

D. Rendered Features vs. Original RADIO Features

Table VI supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of the teacher. It can sharpen the task-relevant scene response after multiview reconstruction: under the same calibrated mask readout it improves both localization and macro mIoU over frame-wise RADIO RGB features, even though the original

TABLE VI
SAME-EVALUATOR COMPARISON BETWEEN ORIGINAL RADIO FEATURES
EXTRACTED FROM RGB FRAMES AND CTF-GS RENDERED FEATURES.
BOTH USE THE SAME FROZEN SIGLIP2 HEAD, SCENE-CATEGORY
SCORING, AND LABEL-FREE THRESHOLD-0.60 PLUS
PEAK-CONNECTED-COMPONENT MASK READOUT.

Scene	RADIO RGB		CTF-GS	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7500	0.4031	0.8214	0.5134
Ramen	0.9014	0.5241	0.9014	0.6249
Teatime	0.8475	0.5429	0.8983	0.6177
Waldo Kitchen	0.7273	0.3688	0.8182	0.5268
Macro	0.8065	0.4597	0.8598	0.5707

TABLE VII
CONTROLLED LERF-OVS COMPONENT ABLATION ON SEED 7. THE
TABLE REPORTS RENDERED-FEATURE LOCACC PER SCENE PLUS MACRO
LOCACC AND mIoU.

Variant	Fig.	Ramen	Tea.	Waldo	LocAcc	mIoU
Full CTF-GS	0.768	0.901	0.898	0.864	0.858	0.485
w/o FGC warm-start	0.696	0.845	0.847	0.818	0.802	0.424
w/o VFA	0.768	0.859	0.915	0.818	0.840	0.480
w/o HGCF	0.768	0.873	0.898	0.818	0.839	0.507
w/o CTR	0.268	0.648	0.661	0.545	0.531	0.260

RADIO RGB feature remains the direct 2D teacher used during training.

Two controls are retained in the supplementary material. A nearest-view RADIO cache without geometric warping reaches only 0.2722 LocAcc / 0.1545 mIoU, and a raw per-Gaussian 1280-D foundation-feature memory reaches 0.5642 / 0.3182 while using 1039.7 MiB mean feature storage. Both trail compact CTF-GS, supporting the claim that learned multiview feature reconstruction is more useful than either a 2D cache or sparse high-dimensional foundation-feature attachment. We additionally consolidate the 2D teacher-vs-CTF-GS evidence across SigLIP2 text grounding and frozen SAM3/DINOv3 task probes in a supplementary table: rendered CTF-GS features win all 6 selected primary downstream metrics. This summary is intentionally phrased as selected downstream feature usability, because several secondary LocAcc/HitRate probes remain stronger for the frame-wise teacher under the same readout.

E. Core Component Ablation

Table VII isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table I, which reports the conservative best per scene, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: the contextual Gaussian field tests compact feature storage, foundation-space reconstruction tests the nonlinear compact feature bottleneck, view-conditioned calibration tests peak and boundary correction, and frozen-head geometry warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with

every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU (0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy on Ramen/Waldo. This explains why some enhancement branches can improve mIoU without preserving LocAcc.

Table VIII consolidates the main quantitative ablation evidence across tasks. The largest gains come from making primitive features queryable through multiview-registration-to-field support and from the foundation-space reconstruction codec. Readout calibrations such as peak-component filtering, feature-only SAM3 boundary refinement, and direct-3D component support are still useful, but their deltas are smaller than the core feature-quality modules. The supplementary evidence records additional variants such as frozen official SAM3 box readout and same-protocol follow-up runs not used in the main tables.

F. Storage Footprint

Table IX closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian and excludes ordinary RGB/geometry attributes. The latent payload is the scale-dependent semantic memory stored per Gaussian, while the feature package additionally counts scene-global heads, CTR/HCD, and VFA/refiner tensors whose fixed overhead is amortized as scene scale grows. The full checkpoint is conservative: it also includes carried Gaussian geometry/RGB tensors. Even with this accounting, the compact representation uses less storage than direct per-Gaussian teacher features on all four LERF scenes, with larger savings on scenes with more Gaussians. Multiview Primitive Registration itself is a training and analysis bridge and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature storage. We therefore use multiview registration as streamed evidence in the main compact-storage claim rather than counting it as part of the stored scene representation. The compression/downstream correlation audit is kept in the supplementary material: checkpoint saving ratio has only small positive correlation with rendered mIoU and negative correlation with Direct3D mIoU, so the paper does not overclaim compression alone as the source of better querying.

G. Adaptor Downstream Probes

Adaptor-consistency sweeps are kept in the supplementary material to preserve main-paper space. They show a consistent localization/region tradeoff: DINO relation and SAM3 region losses can improve thresholded overlap on Ramen, Teatime, and Figurines, but they sometimes move the single heatmap peak outside small LERF polygons. The paper therefore keeps

TABLE VIII
UNIFIED QUANTITATIVE CONTRIBUTION RANKING. EACH ROW COMPARES A REFERENCE AND AN ENABLED VARIANT UNDER THE SAME LOCAL PROTOCOL UNLESS MARKED AS A SUPPLEMENTARY CONTROL. SCORE IS THE MEAN POSITIVE DELTA OVER THE LISTED METRICS; SECONDARY REGRESSIONS ARE KEPT IN THE METRIC-DELTA COLUMN.

Contribution	Task	Reference → Variant	Metric deltas	Score	Status
MPR-to-field primitive feature registration	LERF Direct3D	raw Gaussian-center score - ζ registered compact primitive score	mIoU: 0.0460- ζ 0.4120 (+0.3660); Acc@0.25: 0.0710- ζ 0.5880 (+0.5170)	0.442	core direct-field evidence
Foundation-space reconstruction codec	LERF rendered architecture	w/o codec - ζ full CTF-GS seed-7 model	LocAcc: 0.5310- ζ 0.8580 (+0.3270); mIoU: 0.2600- ζ 0.4850 (+0.2250)	0.276	core architecture
Final rendered-view readout vs frame-wise RADIO	2D foundation-feature usability	frame-wise RADIO teacher - ζ CTF-GS rendered field + feature-only boundary readout	LocAcc: 0.7985- ζ 0.8598 (+0.0613); mIoU: 0.4634- ζ 0.5889 (+0.1255)	0.093	main claim evidence
Frozen-head geometry warm-start	LERF rendered architecture	w/o geometry warm-start - ζ full CTF-GS seed-7 model	LocAcc: 0.8020- ζ 0.8580 (+0.0560); mIoU: 0.4240- ζ 0.4850 (+0.0610)	0.058	core architecture
Direct3D prompt ensemble + support calibration	LERF Direct3D	strict single-prompt pure one-map - ζ score-component guarded compact readout	mIoU: 0.4489- ζ 0.5014 (+0.0525); Acc@0.25: 0.6724- ζ 0.7044 (+0.0321); Boundary-F: 0.6124- ζ 0.6305 (+0.0181)	0.034	main compact direct readout
DINOv3 dense matching feature readout	2D frozen-head downstream	frame-wise RADIO teacher - ζ CTF-GS rendered field	Mean score: 0.8547- ζ 0.9048 (+0.0501); HitRate: 0.5723- ζ 0.5396 (-0.0327)	0.025	downstream readout evidence
SAM3 point-prompt feature readout	2D frozen-head downstream	frame-wise RADIO teacher - ζ CTF-GS rendered field	mIoU: 0.3700- ζ 0.4173 (+0.0473); LocAcc: 1.0000- ζ 1.0000 (+0.0000)	0.024	downstream readout evidence
Peak-component rendered mask readout	LERF rendered-view grounding	threshold 0.60 mask - ζ threshold 0.60 + peak component	mIoU: 0.5243- ζ 0.5707 (+0.0464); LocAcc: 0.8712- ζ 0.8598 (-0.0114)	0.023	readout calibration

TABLE IX
STORAGE FOOTPRINT ACCOUNTING. DIRECT STORES ONE 1280-D FP16 TEACHER FEATURE PER GAUSSIAN. LATENT IS THE COMPACT PER-GAUSSIAN SEMANTIC PAYLOAD. FEATURE PACKAGE ADDS SCENE-GLOBAL HEADS, CTR/HCD, AND VFA/REFINER TENSORS; THESE FIXED TENSORS ARE AMORTIZED AS SCENE SCALE GROWS. FULL CHECKPOINT ADDITIONALLY COUNTS CARRIED 3DGS GEOMETRY/RGB STATE. LOWER IS BETTER FOR STORAGE; HIGHER SAVING RATIOS ARE BETTER.

Scene	#G	Direct MiB	Latent MiB	Save	Feature pkg. MiB	Save	Full ckpt. MiB	Save
Figurines	168,791	412.1	20.6	20.0×	199.0	2.07×	237.0	1.74×
Ramen	382,687	934.3	46.7	20.0×	225.1	4.15×	311.2	3.00×
Teatime	460,157	1123.4	56.2	20.0×	234.5	4.79×	338.1	3.32×
Waldo Kitchen	691,728	1688.8	84.4	20.0×	262.8	6.43×	418.5	4.04×

the calibrated LERF readout as the main grounding result and reports the prompt-constrained downstream probes below as the compact evidence for DINO/SAM feature usability.

The broader unconstrained variants and cross-view DINO controls remain in the supplementary material. They are useful for understanding outliers, but the prompt-constrained tasks better match how frozen SAM/DINO heads are used downstream.

Table X separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over the frame-wise teacher for point prompts, box prompts, and mask propagation, while matching the teacher’s point-prompt localization. Second, using the same rendered field for DINO support and SAM-adaptor boundary refinement makes DINO propagation positive under the same evaluator: rendered features reach 0.7872/0.4677 LocAcc/mIoU versus 0.7660/0.4606 for the frame-wise teacher under the same multi-head readout. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants eval-

TABLE X
PROMPTABLE SAM3-ADAPTOR AND DENSE DINOv3-ADAPTOR TASKS ON LERF-OVS. THESE PROBES USE THE SAME ANNOTATIONS AS THE GROUNDING BENCHMARK BUT REPLACE TEXT QUERIES WITH ADAPTOR-SPACE PROMPTS OR SOURCE-TARGET CORRESPONDENCES. SAM3 POINT/BOX PROMPTS ARE PROMPT-CONSTRAINED, MASK PROPAGATION USES SOURCE-VIEW SUPPORT ONLY, AND NO EXTERNAL SAM3 MASK DECODER IS CALLED. DINO MASK PROPAGATION USES SOURCE-BACKGROUND CONTRAST 1.10 PLUS LABEL-FREE MULTI-HEAD READOUT: TOP- k FOREGROUND POOLING, $2.0\times$ PROPAGATED-AREA SCALING, PEAK-COMPONENT CLEANUP, AND SAM-ADAPTOR BOUNDARY REFINEMENT.

Task	Teacher		Rendered	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4173
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6596	0.3756
DINOv3 dense matching	0.5723	0.8547	0.5396	0.9048
DINOv3 mask propagation + SAM-boundary readout	0.7660	0.4606	0.7872	0.4677

uated next. We also evaluate a mutual-correspondence plus homography-RANSAC control for qualitative DINO matching; it removes many outlier matches and raises rendered dense-match similarity to 0.9277, but it does not improve the same robust DINO only mask-propagation mIoU beyond the teacher. We therefore keep RANSAC as a visualization control and use the multi-head DINO-support/SAM-boundary readout for the quantitative frozen-head table.

H. ScanNet Direct Point-Query Transfer

Table XI evaluates cross-domain feature usability on the VALA/OpenGaFF ScanNet-8 scene subset: scene0000_00, scene0062_00, scene0070_00, scene0097_00, scene0140_00, scene0347_00, scene0400_00, and scene0590_00. This follows the OpenGaFF/VALA direct point-query setting with every-20-frame ScanNet training splits and mIoU/mAcc evaluation on GT semantic point clouds. This is not presented as a full ScanNet semantic

TABLE XI

SCANNet-v2 OPEN-VOCABULARY POINT-CLOUD UNDERSTANDING UNDER THE OPENGAFF/VALA PROTOCOL. EXTERNAL BASELINE ROWS ARE COPIED FROM THE OPENGAFF V2 PAPER TABLE AND REPORTED IN PERCENT; THE OPENGAFF METHOD ROW IS INTENTIONALLY OMITTED. THE CTF-GS ROW IS OUR LOCAL VALA/OPENGAFF-8 EVALUATION ON THE SAME EIGHT SCENE IDS.

Method	19 mIoU	19 mAcc	15 mIoU	15 mAcc	10 mIoU	10 mAcc
LangSplat	2.45	8.59	3.45	13.21	6.48	21.89
LangSplatV2	14.75	25.47	17.09	35.68	22.83	41.52
OpenGaussian	27.73	42.01	29.67	46.15	39.93	57.34
Dr. Splat	29.31	47.68	33.25	54.33	44.19	65.19
OccamLGS	31.93	48.93	34.25	53.71	45.16	64.39
VALA	32.11	50.05	35.10	54.77	46.21	65.61
CTF-GS DINO-CV contextual kNN + spatial propagation	38.06	61.29	38.71	63.15	47.11	72.00

TABLE XII

SCANNet VALA8 CATEGORY STABILITY FOR THE CTF-GS DINO-CV CONTEXTUAL KNN READOUT WITH SPATIAL LOGIT PROPAGATION. STATISTICS ARE PER-CLASS IOU ACROSS SCENES WHERE THE CLASS APPEARS; “RANGE” IS MIN-MAX IOU.

Split	Weakest class	Scenes	Mean IoU	Range	Most unstable class	Std IoU	Range
19	picture	3	0.018	0.000-0.053	refrigerator	0.297	0.004-0.827
15	table	4	0.126	0.000-0.374	door	0.238	0.015-0.786
10	toilet	2	0.216	0.114-0.318	window	0.244	0.180-0.867

segmentation leaderboard result; it is a split-aligned direct probe of the learned feature field. The CTF-GS result uses the cross-view DINO-consistent compact field with a label-free contextual kNN point readout ($k = 16, 80$ candidate Gaussians), single-template text prompts, and scene-mean logit calibration at $\alpha = 0.45$. A label-free spatial logit propagation step averages class logits over 12 nearest 3D neighbours before the final argmax. It reaches 0.3806/0.6129, 0.3871/0.6315, and 0.4711/0.7200 mIoU/mAcc on the 19/15/10-class splits. The external results are copied from the OpenGaFF paper but the OpenGaFF method result is omitted, so the table is used as compact baseline context rather than as a head-to-head OpenGaFF comparison. Increasing the calibration scale to 0.75 raises 10-class mIoU to 0.4612 but lowers the 19-/15-class balance; alias prompts and the stricter joint 2D/3D training loss are kept in the supplementary material.

Table XII summarizes the main ScanNet failure modes for the CTF-GS result. The weakest classes are either small decorative regions such as *picture*, broad cluttered surfaces such as *table*, or classes with sparse scene support such as *toilet*; unstable classes such as *door*, *window*, and *refrigerator* vary strongly with view coverage and object extent.

I. Efficiency and Cost

Table XIII reports lower-is-better single-query latency derived from frozen profile artifacts, separate from storage footprint accounting. The reported units match each protocol: rendered-view LERF uses view-query latency, direct 3D LERF uses object-query latency after query-select-render mask generation, and ScanNet uses class-query latency. These values are conservative because the frozen profiles include evaluator work and, for rendered-view LERF, visualization/teacher branches; clean warm-GPU microbenchmarks are kept out of the main

TABLE XIII

SINGLE-QUERY LATENCY EVIDENCE. LOWER IS BETTER. VALUES ARE DERIVED FROM FROZEN PROFILE ARTIFACTS AND ARE CONSERVATIVE BECAUSE THE PROFILED COMMANDS INCLUDE EVALUATOR OVERHEAD AND, FOR LERF RENDERED-VIEW RUNS, VISUALIZATION/TEACHER BRANCHES. DIRECT 3D LATENCY INCLUDES QUERY-SELECT-RENDER MASK GENERATION.

Task	Unit	Queries	Latency ms/query	Peak VRAM MiB
LERF rendered-view OVS	view-query	356	350.5	2076
LERF direct 3D OVS	object-query	208	2367.7	–
ScanNet point query	class-query	389	387.9	1666

claim until run under a single instrumentation harness. Training throughput remains supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types.

J. Qualitative Results

The main qualitative comparison in Fig. 2 follows the LERF-OVS taxonomy used in recent open-vocabulary Gaussian-field work: rendered 2D querying and direct 3D primitive querying are shown side by side on the same annotated view and text query. The 2D prior panels use our local LangSplat reproduction, while the 3D prior panels use our local Dr. Splat reproduction. The CTF-GS 3D panels use the deployed compact field with the fixed Support-Calibrated Primitive Readout; they do not read a multiview-registration feature cache and do not call the official RGB SAM3 readout at evaluation time.

Fig. 3 gives the corresponding direct point-query visualization on ScanNet. We use binary class-query point clouds rather than full semantic coloring, because the evaluation protocol asks whether a frozen text query retrieves a target class under the VALA/OpenGaFF split.

Fig. 4 visualizes the direct-3D support-calibration ablation on small or fragmented LERF queries. The base compact readout uses primitive scores from the compact field, while the final readout adds the prompt ensemble and label-free component-support calibration used by the deployed compact direct readout.

Fig. 5 isolates the rendered-view boundary calibration used by the feature-only SAM3-adaptor readout. Unlike the frozen

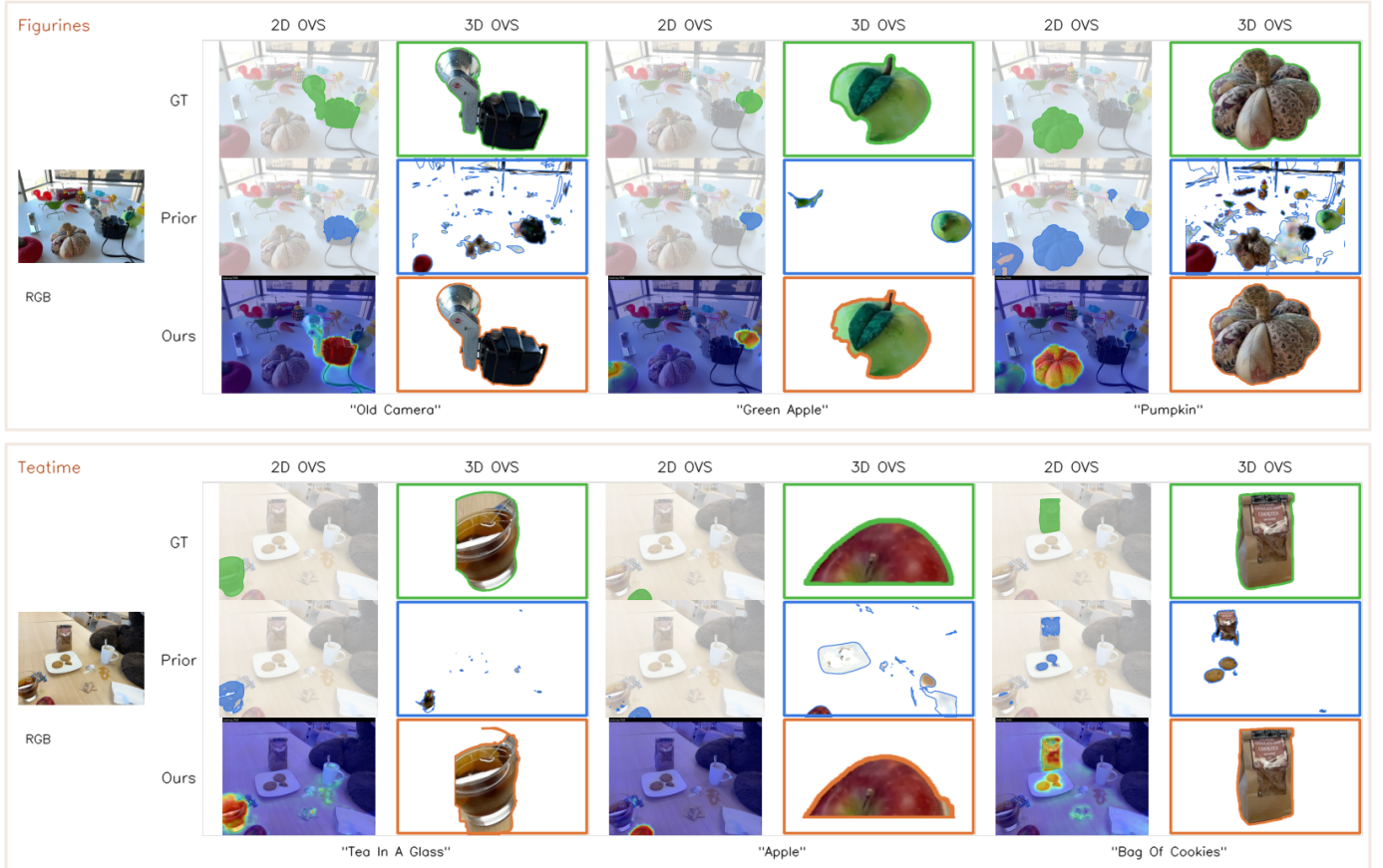


Fig. 2. Qualitative comparison of 2D and 3D open-vocabulary querying on LERF-OVS. For each text query, the 2D columns visualize rendered-view mask/heatmap outputs on RGB, while the 3D columns visualize direct Gaussian-level selection rendered on a white background with local zoom. The prior result uses reproduced local baselines: LangSplat for 2D OVS and Dr. Splat for 3D OVS. CTF-GS uses one compact foundation-feature Gaussian map for both readouts; the 3D panels do not use a multiview-registration cache or official RGB SAM3 readout.

official SAM3 diagnostic, this readout is conditioned on reconstructed CTF-GS/RADIO features and prompt embeddings; it does not call the official RGB SAM3 decoder at evaluation time.

Fig. 6 compares reconstructed scene features with frame-wise RADIO features under the frozen downstream probes used in Table X. These examples visualize the same claim as the quantitative probe table: the compact scene field is not merely a compressed copy of a single view, but a multiview feature memory that can produce more stable downstream responses.

Additional rendered heatmap examples, alpha/depth boundary cases, and extended SAM/DINO-adaptor controls are kept in the supplementary material so the main paper focuses on one representative figure per claim.

K. Failure Analysis

Table XIV turns the qualitative small-object observation into a quantitative failure analysis. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a plausible object region while

TABLE XIV
REPRESENTATIVE FRAGILE LERF-OVS CATEGORIES FROM THE FROZEN RENDERED-FEATURE EVALUATION. LOW LOCACC WITH NONZERO mIoU HIGHLIGHTS THE PEAK-VS-REGION TRADE-OFF.

Scene	Category	LocAcc	mIoU
Figurines	bag	0.00	0.00
Figurines	miffy	0.00	0.01
Teatime	dall-e brand	0.00	0.01
Figurines	jake	0.00	0.02
Waldo Kitchen	ketchup	0.00	0.03
Waldo Kitchen	spatula	0.00	0.03
Ramen	corn	0.00	0.10
Figurines	red toy chair	0.00	0.13

the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

V. DISCUSSION

a) *Why feature reconstruction matters.*: The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localization. The ScanNet results suggest that the same representation can be probed

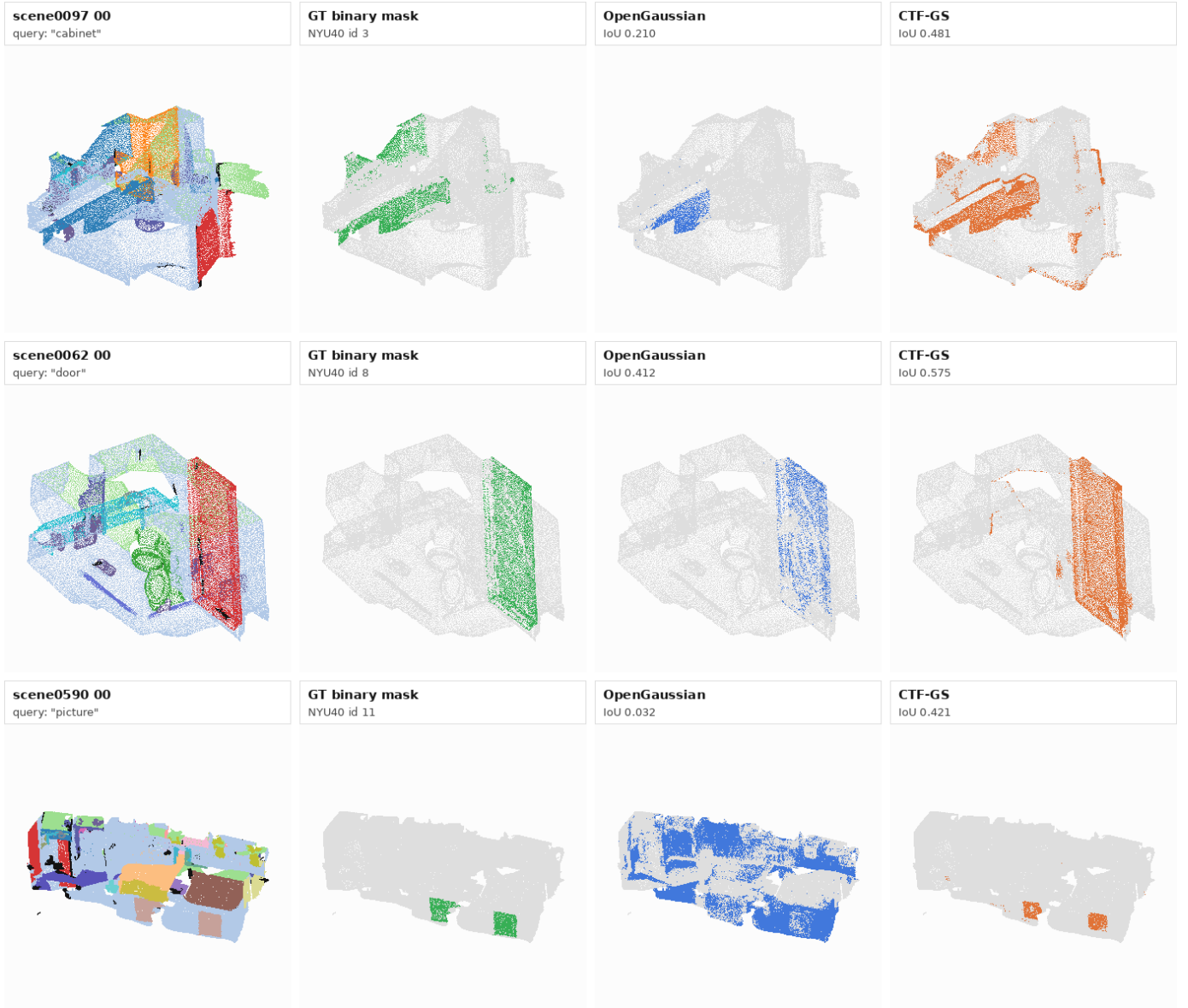


Fig. 3. ScanNet open-vocabulary 3D query qualitative comparison. Each row shows a binary text-class query on a VALA/OpenGaff-8 scene: scene overview, GT class points, our local OpenGaussian reproduction, and CTF-GS direct point-query prediction. Gray points are non-query classes.

outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying, Multiview Primitive Registration provides an auditable training bridge for primitive evidence, and support-calibrated readout is needed to convert primitive scores into stable object masks. Bare score-threshold selection is still weaker than object-aware grouping on fragmented scenes, so we distinguish primitive scoring quality from boundary and support readout quality.

b) External-baseline scope.: The LERF, LangSplat, and LEGaussians comparison results are traced to official paper or supplementary tables. They still come from cross-paper protocols, so we treat them as source-anchored context. A strict ranking claim would require rerunning those baselines under the same evaluator and annotations used by RADIO-GS.

c) Why the compact field can outperform its teacher.: CTF-GS is trained from frame-wise RADIO features, but its rendered readout is not restricted to a single frame. Multiview reconstruction, visibility-aware regularization, and support-calibrated readouts can suppress view-specific noise and recover a scene-consistent response. This explains why the rendered field can outperform frame-wise RADIO under the same frozen SigLIP2, SAM3-adaptor, and DINOv3 probe tasks, while still being limited by teacher resolution and annotation ambiguity on small objects.

VI. LIMITATIONS

CTF-GS depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because the teacher feature resolution and rendered feature alignment limit heatmap sharpness. The ScanNet protocol is a direct point-query feature

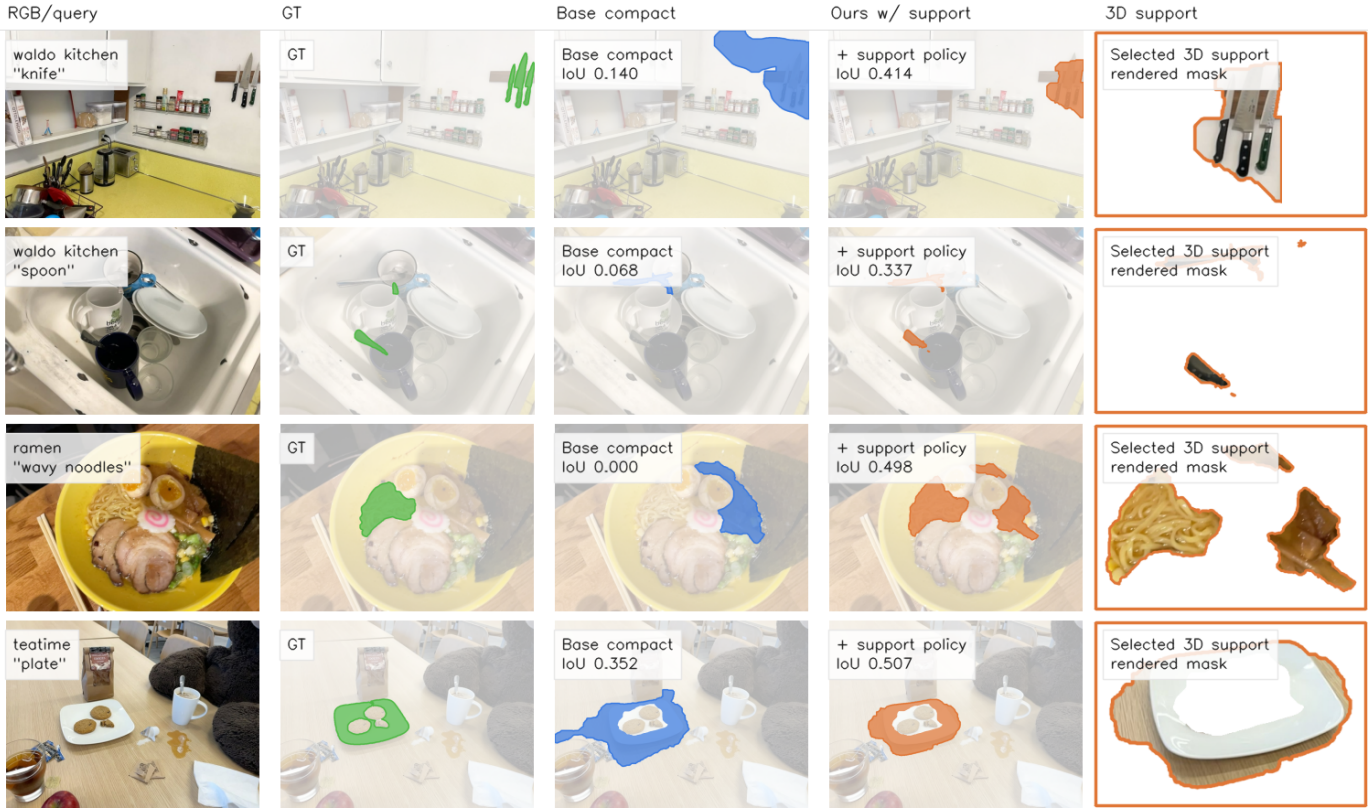


Fig. 4. Qualitative ablation of the compact direct-3D support-calibrated readout. The readout recovers small or fragmented object support without a multiview-registration feature cache or official RGB SAM3 readout; rendering is used only to visualize and evaluate the selected 3D primitives.

probe, not a replacement for a full standardized segmentation benchmark. The registered LERF direct 3D object-selection analysis substantially improves primitive querying and exceeds the OpenGaussian official macro reference under the reported OpenGaussian-style context, but the external results are not locally rerun under one evaluator; universal primitive-level ranking claims would require those reruns. The deployed compact direct-3D readout uses label-free color-edge and score-component support calibration. It avoids multiview-registration cache inference and official RGB SAM decoding, but it is not the strict no-RGB ablation; that stricter readout remains lower at 0.457 macro mIoU. The frozen official SAM3 box result is therefore kept as a post-selection boundary readout rather than the core direct-field claim. The storage table reports checkpoint footprint rather than deployed runtime allocation; a deployment-oriented memory study would separately account for feature-field parameters, rendering buffers, and optional downstream heads.

VII. CONCLUSION

CTF-GS turns a 3D Gaussian scene into a reusable foundation-feature memory. By reconstructing frozen RADIO features rather than training a scene-specific classifier, it supports open-vocabulary grounding and cross-domain feature queries from novel views. The experiments provide strong LERF-OVS same-protocol results, split-aligned ScanNet direct-query evidence, explicit storage and runtime pro-

files, failure analysis, and source-anchored external comparison protocol boundaries.

REFERENCES

- [1] J. Kerr, C. M. Kim, K. Goldberg, A. Kanazawa, and M. Tancik, “Lerf: Language embedded radiance fields,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [2] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister, “Langsplat: 3d language gaussian splatting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [3] J.-C. Shi, M. Wang, H.-B. Duan, and S.-H. Guan, “Language embedded 3d gaussians for open-vocabulary scene understanding,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [4] Y. Wu, J. Meng, H. Li, C. Wu, Y. Shi, X. Cheng, C. Zhao, H. Feng, E. Ding, J. Wang, and J. Zhang, “Opengaussian: Towards point-level 3d gaussian-based open vocabulary understanding,” *arXiv preprint arXiv:2406.02058*, 2024.
- [5] J.-S. Kim, G. Kim, Y.-J. Kim, Y.-C. F. Wang, J. Choe, and T.-H. Oh, “Dr. splat: Directly referring 3d gaussian splatting via direct language embedding registration,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [6] H. Li, Y. Wu, J. Meng, Q. Gao, Z. Zhang, R. Wang, and J. Zhang, “Instancegaussian: Appearance-semantic joint gaussian representation for 3d instance-level perception,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [7] J. Piekenbrinck, C. Schmidt, A. Hermans, N. Vaskevicius, T. Linder, and B. Leibe, “Opensplat3d: Open-vocabulary 3d instance segmentation using gaussian splatting,” *arXiv preprint arXiv:2506.07697*, 2025.
- [8] T. Huang, R. Chen, D. Hu, F. Huang, M. Gong, and T. Liu, “Openins-gaussian: Open-vocabulary instance gaussian segmentation with context-aware cross-view fusion,” *arXiv preprint arXiv:2510.18253*, 2025.
- [9] B. L. Nguyen, K. Nguyen, S. Sridharan, C. Fookes, and P. Moghadam, “Ilov3splat: Instance-level open-vocabulary 3d scene understanding in gaussian splatting,” *arXiv preprint arXiv:2605.04506*, 2026.

Rendered-view boundary calibration: coarse heatmap support $\rightarrow$ feature-only boundary mask

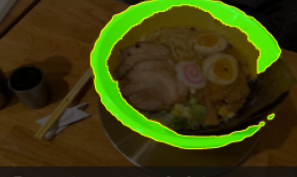
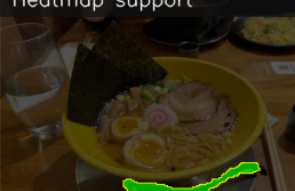
Query	GT	Heatmap support	Feature-only boundary
ramen "bowl" +0.20 IoU		 Coarse IoU 0.44	 Boundary IoU 0.64
ramen "bowl" +0.16 IoU		 Coarse IoU 0.45	 Boundary IoU 0.61
ramen "onion segments" +0.14 IoU		 Coarse IoU 0.43	 Boundary IoU 0.57
ramen "spoon" +0.11 IoU		 Coarse IoU 0.57	 Boundary IoU 0.68

Fig. 5. Rendered-view boundary calibration qualitative examples. The coarse heatmap support is converted into a feature-only boundary mask using the prompt-conditioned SAM3-adaptor head. The examples show per-query IoU gains without invoking an external RGB SAM3 decoder.

- [10] S. Liang, S. Wang, K. Li, M. Niemeyer, S. Gasperini, H. P. A. Lensch, N. Navab, and F. Tombari, "Supergseg: Open-vocabulary 3d segmentation with structured super-gaussians," *arXiv preprint arXiv:2412.10231*, 2026.
- [11] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Transactions on Graphics*, 2023.
- [12] M. Ranzinger, G. Heinrich, J. Kautz, and P. Molchanov, "Am-radio: Agglomerative vision foundation model – reduce all domains into one," *arXiv preprint arXiv:2312.06709*, 2024.
- [13] NVIDIA, "C-radiv4: Tech report," *arXiv preprint arXiv:2601.17237*, 2026.
- [14] X. Zuo, P. Samangouei, Y. Zhou, Y. Di, and M. Li, "Fmgs: Foundation model embedded 3d gaussian splatting for holistic 3d scene understanding," *arXiv preprint arXiv:2401.01970*, 2024.
- [15] J. Cen, J. Fang, C. Yang, L. Xie, X. Zhang, W. Shen, and Q. Tian, "Segment any 3d gaussians," *arXiv preprint arXiv:2312.00860*, 2023.
- [16] Y.-J. Chiou, W.-T. Cheng, and Y.-F. Yang, "Profuse: Efficient cross-view context fusion for open-vocabulary 3d gaussian splatting," *arXiv preprint arXiv:2601.04754*, 2026.

Reconstructed scene features vs. frame-wise RADIO across frozen-head tasks

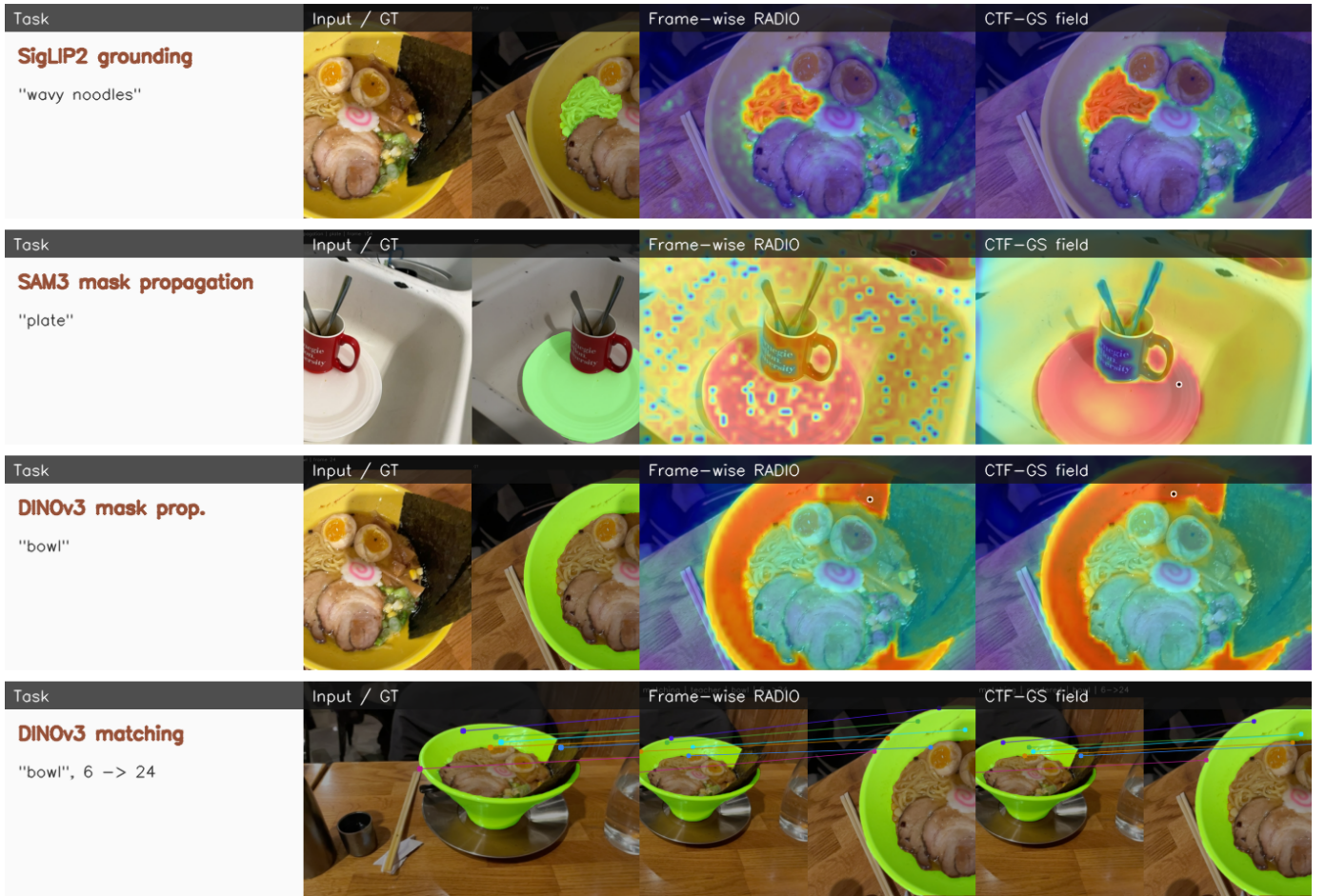


Fig. 6. Reconstructed scene features versus frame-wise RADIO under frozen-head tasks. Columns compare the task input/GT, frame-wise RADIO response, and CTF-GS reconstructed field response for SigLIP2 grounding, SAM3-adaptor mask propagation, DINOv3 mask propagation, and DINOv3 dense matching.